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Micro injection flooring repair assembly and method

Abstract

A micro injection assembly for the repair of floors that includes an adapter body operably configured to couple with a tube having adhesive and defining an internal channel, a hub selectively removably coupled to the adapter body and having a rigid needle member coupled thereto and recessed within the hub, wherein the rigid needle member projects outwardly from the second end of the hub in a linear orientation, has a blunt distal end, and defines a needle channel spanning the length of the needle member. The assembly also includes a cap selectively removably couplable to the adapter body in a watertight configuration and with the cap disposable in an overlapping configuration with the adapter body. The assembly is operable to effectuate emission of an adhesive material from a tube through the adapter body, hub, and needle member to spaces between a flooring material and subfloor.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9151066	12/2014	Hilton	N/A	E04G 23/0285
2010/0155424	12/2009	Tsai	222/326	B65D 81/325
2015/0158050	12/2014	Atwell	222/420	B65D 35/38

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-2010041075	12/2009	WO	B05C 17/00513
WO-2022063703	12/2021	WO	B05C 17/00513

OTHER PUBLICATIONS

Hypodermic Needle Gauge Chart. Medical Tube Technology, Inc. © 2004.

“http://www.medtube.com/hypo_chrt.htm”. Accessed via the Wayback Machine on Feb. 27, 2025 at “https://web.archive.org/web/20041010031333/http://www.medtube.com/hypo_chrt.htm”. (Year: 2004). cited by examiner

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates generally to flooring repair, and more particularly, to an assembly and method of injecting adhesive into flooring including tile, marble, stone, wood, engineered, laminate, vinyl, etc. and/or very narrow joints or tight spaces or floors with very narrow or no grout lines.

BACKGROUND OF THE INVENTION

(2) Traditional methods for repairing a floor where the flooring material has become separated from the subfloor, such as a concrete slab, usually involve removing the flooring material from the subfloor and cleaning both the subfloor and the flooring material, then reinstalling the flooring material. In some situations, the flooring material cannot be reinstalled, which then requires finding matching replacement flooring material or replacing all the flooring material, which can be difficult or impossible to do.

(3) More recent methods for repairing a floor where the flooring material has become separated from the subfloor include delivering a liquid adhesive between the flooring material and the subfloor via a few drilled holes in the floor while leaving the flooring material in place. The liquid adhesive is then allowed to set, thereby adhering the flooring material to the subfloor. A method like this is disclosed in Applicant's prior patent, U.S. Pat. No. 9,151,066, entitled an "apparatus and method to fix a floor", which is incorporated in its entirety herein by reference. This prior art however has the disadvantage of a wider injection nozzle tip (approximately 1/8-inch diameter), preventing efficient and successful use of the assembly in cases of very narrow or no grout line applications, or in marble, tile, stone, wood, laminate, vinyl, etc.

(4) Therefore, a need exists to overcome the problems with the prior art as discussed above.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views and which together with the detailed description below are incorporated in and form part of the specification, serve to further illustrate various embodiments and explain various principles and advantages all in accordance with the present invention.

(2) FIG. 1 is a front perspective view of an adaptor of a micro injection assembly with an injection needle therein, in accordance with an exemplary embodiment of the micro injection assembly;

(3) FIG. 2 is a sectional view of the micro injection assembly in FIG. 1 along section line 2-2, in accordance with the present invention;

(4) FIG. 3 is a front perspective view of the micro injection assembly according to FIG. 1 with a resealable cap thereon, in accordance with the present invention;

(5) FIG. 4 is a top perspective view of a liquid adhesive being applied using the micro injection assembly and method, in accordance with the present invention;

(6) FIG. 5 is a front elevational view of several exemplary embodiments of an injection needle of the micro injection assembly, in accordance with the present invention; and

(7) FIG. 6 depicts an exemplary process flow diagram showing a method of repairing loose flooring in accordance with one embodiment of the present invention.

SUMMARY OF THE INVENTION

(8) The invention provides a micro injection flooring repair assembly and method that overcomes the hereinafore-mentioned disadvantages of the heretofore-known devices and methods of this general type and that provides a more efficient applicator of a liquid adhesive on broken, separated, squeaky, hollow or loose floors. The present invention is operably configured to fix loose wood flooring, marble floors, tile floors, stone floors, laminate, etc., as well as floors with no grout lines like marble or where existing tips are still too big for the application. This allows for many more additional and versatile ways to inject adhesives into the floor.

(9) With the foregoing and other objects in view, there is provided, in accordance with the invention, a micro injection assembly for the repair of floors that includes an adapter body having a first end, a second end opposing the first end of the adapter body, a recessed internal wall surface

defining a threaded configuration operably configured to couple with a threaded tube, and defining an internal channel spanning from the first end of the adapter body to the second end of the adapter body, a hub selectively removably coupled to the adapter body, having a first end disposed in a directly adjacent configuration with the second end of the adapter body, having a second end opposing the first end of the hub, and having a rigid needle member coupled thereto and recessed within the hub, wherein the rigid needle member projects outwardly from the second end of the hub in a linear orientation, has a blunt distal end, and defines a needle channel spanning from the blunt distal end to the first end of the hub, wherein the needle channel is in fluid communication with the internal channel. The assembly also includes a cap selectively removably couplable to the adapter body in a watertight configuration with the cap disposable in an overlapping configuration with the second end of the adapter body, i.e., shaped and sized to couple with the adapter body.

(10) In accordance with another feature, an embodiment of the present invention includes the adapter body having a lower section having the first end of the adapter body and the recessed internal wall surface and an upper section adjacent to the lower section, having the second end of the adapter body, and tapered in diameter relative to a diameter of the lower section of the adapter body.

(11) In accordance with yet another feature, an embodiment of the present invention also includes the upper section having a spout surrounded by a recessed annular channel, the recessed annular channel with the hub disposed therein.

(12) In accordance with a further feature of the present invention, the lower section and the upper section are of a monolithic structure.

(13) In accordance with an additional feature of the present invention, the rigid needle member is of a slender configuration.

(14) In accordance with a further feature of the present invention, the rigid needle member ranges in dimension from 12-30 gauge.

(15) In accordance with yet another feature, an embodiment of the present invention also includes the adapter body having a spout surrounded by a recessed annular channel, the recessed annular channel with the hub disposed therein.

(16) In accordance with a further feature, an embodiment of the present invention also includes an overall hub-needle length separating the first end of the hub and the distal end of the needle member and a needle member length separating the second end of the hub and the distal end of the needle member, wherein the overall hub-needle length has a length variance ranging from 0.5-1.5 in. relative to the needle member length.

(17) The present invention is also embodied in a method of repairing loose flooring that includes the steps of providing an adapter body with a threaded configuration and an internal channel therethrough, a hub with a rigid needle member coupled thereto, of a slender configuration, having, defining a needle channel therethrough, and having a distal end defining an opening, and a cap, coupling the threaded configuration of the adapter body to a threaded configuration of a tube filled with an adhesive material and coupling the hub to the adapter body in a watertight configuration, inserting the rigid needle member into an opening adjacent to a piece of flooring material loosely coupled to a subfloor, modulating the tube to transport the adhesive material from the tube through the internal channel, through the needle channel, out through the opening of the distal end of the rigid needle member, and between the flooring material and subfloor, and uncoupling the hub from the adapter body adapter body and coupling the cap to the adapter body in a watertight configuration.

(18) In accordance with a further feature, an embodiment of the present invention also includes coupling the cap to the adapter body in a hermetically sealed configuration.

(19) In accordance with an exemplary feature, an embodiment of the present invention also includes coupling the hub to the adapter body in directly adjacent configuration.

(20) In accordance with an additional feature, an embodiment of the present invention also includes

providing the adapter body with a spout surrounded by a recessed annular channel and coupling the hub to the adapter body until inserted within the recessed annular channel.

(21) In accordance with yet another feature, an embodiment of the present invention also includes threadedly coupling the hub to the adapter body and providing the rigid needle member projecting from the hub in a linear orientation and having a blunt distal end.

(22) In accordance with an exemplary feature, an embodiment of the present invention also includes providing the adapter body with a lower section having a first end of the adapter body and a recessed internal wall surface and an upper section adjacent to the lower section, having a second end of the adapter body located thereon, and tapered in diameter relative to a diameter of the lower section of the adapter body and coupling the recessed internal wall surface, with the threaded configuration thereon, to the adapter body to the tube.

(23) In accordance with a further feature, an embodiment of the present invention also includes providing the rigid needle member that ranges in dimension from 12-30 gauge.

(24) Although the invention is illustrated and described herein as embodied in a micro injection flooring repair assembly and method, it is, nevertheless, not intended to be limited to the details shown because various modifications and structural changes may be made therein without departing from the spirit of the invention and within the scope and range of equivalents of the claims. Additionally, well-known elements of exemplary embodiments of the invention will not be described in detail or will be omitted so as not to obscure the relevant details of the invention.

(25) Other features that are considered as characteristic for the invention are set forth in the appended claims. As required, detailed embodiments of the present invention are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the invention, which can be embodied in various forms. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one of ordinary skill in the art to variously employ the present invention in virtually any appropriately detailed structure. Further, the terms and phrases used herein are not intended to be limiting; but rather, to provide an understandable description of the invention. Unless otherwise indicated by Applicant, the figures of the drawings are not necessarily drawn to scale.

(26) Before the present invention is disclosed and described, it is to be understood that the terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. The terms “a” or “an,” as used herein, are defined as one or more than one. The term “plurality,” as used herein, is defined as two or more than two. The term “another,” as used herein, is defined as at least a second or more. The terms “including” and/or “having,” as used herein, are defined as comprising (i.e., open language). The term “coupled,” as used herein, is defined as connected, although not necessarily directly, and not necessarily mechanically. The term “providing” is defined herein in its broadest sense, e.g., bringing/coming into physical existence, making available, and/or supplying to someone or something, in whole or in multiple parts at once or over a period of time. Also, for purposes of description herein, the terms “upper”, “lower”, “left,” “rear,” “right,” “front,” “vertical,” “horizontal,” and derivatives thereof relate to the invention as oriented in the figures and is not to be construed as limiting any feature to be a particular orientation, as said orientation may be changed based on the user's perspective of the device. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, brief summary or the following detailed description.

(27) As used herein, the terms “about” or “approximately” apply to all numeric values, whether or not explicitly indicated. These terms generally refer to a range of numbers that one of skill in the art would consider equivalent to the recited values (i.e., having the same function or result). In many instances these terms may include numbers that are rounded to the nearest significant figure. In this document, the term “longitudinal” should be understood to mean in a direction

corresponding to an elongated direction of the needle member and/or adhesive tube to which the needle member is attached, where applicable.

DETAILED DESCRIPTION OF THE INVENTION

(28) While the specification concludes with claims defining the features of the invention that are regarded as novel, it is believed that the invention will be better understood from a consideration of the following description in conjunction with the drawing figures, in which like reference numerals are carried forward.

(29) With reference to FIG. 1, the present invention provides a micro injection flooring repair assembly and method **100** (referred to as “assembly **100**” hereinafter for brevity) for repairing loose or broken flooring in particularly narrow or tight spaces. Embodiments of the invention provide an assembly **100** and method that may be applied in connection with various types of flooring (e.g., wood, laminate, marble, etc.).

(30) The assembly **100** is operably configured to be used for the repair of floors where a flooring material **300** (e.g., wood, tile, marble, etc.) has become separated from a subfloor **302** (e.g., wood or concrete, etc.) and a void is created between the flooring material **300** and the subfloor **302**. The assembly **100** comprises an adaptor **102** comprising a body **104** having a first end **106**, a second end **108** opposite the first end **106** and selectively resealable with a resealable cap **400**. The body **104** can be seen defining an internal channel **200** spanning between the first end **106** and the second end **108** of the body **104**, wherein the internal channel **200** is shaped, sized, and configured (with the first end **106**) to receive and transport a liquid adhesive from an upper aperture **202** defined by a mouth **112** of an adhesive tube **114**.

(31) The body **104** can be seen having adjacent tapering wall sections, wherein the upper section of can be seen defining the second end **108** and the lower section can be seen defining the first end **106**. The upper section has an outer diameter less than the outer diameter of the lower section to facilitate in enabling the adapter **102** to reach smaller areas. The adapter is preferably cylindrical and has a diameter of approximately 0.5-2 in. The upper section of the body **104** can be seen defining a recessed annular channel shaped to receive a hub **116** having a substantially rigid (for brevity, referred to as “rigid”) needle member **110** coupled thereto. More specifically, the body **104** can be seen defining a spout **208** at or defining (at least in part) the second end **108** and that is surrounded by the annular channel. The spout **208** may define an enclosure spout aperture in fluid communication with the internal channel **200**, wherein the hub **116** is configured to selectively couple with and surround the spout of the when the user desires to inject and emit adhesive.

(32) The hub **116** and/or needle member **110** can be seen defining an elongated needle channel **204** that spans from the opening of the spout **208** to the distal end, or free end, **206**. In one embodiment, the needle channel **204** is linear (like the structure of the needle member **110**) and, when not connected to the spout **208**, spans from a first end **210** of the hub **116** to the distal end **206** of the needle member **110**. In one embodiment, the needle member **110** may be permanently coupled or fastened to the hub **116** with, for example, adhesive, to reduce possibility of failure when inserting, for example, the needle member **110** into an opening **304** defined by two opposing pieces of flooring **302**. In other embodiments, the needle member **110** may be selectively removable from the hub **116** with one or more fasteners or a tongue-and-groove coupling configuration.

(33) In one embodiment, the second end **108** of the adapter body **104** is configured to receive the hub **116** of the injection needle **110** by selectively screwing the hub **116** to the adapter body **104**. Said another way, the hub **116** and/or needle member **110** may be selectively removably couplable to the adapter body **104** with a tongue-and-groove configuration, which includes a threaded configuration. In preferred embodiments, this coupling configuration is a watertight configuration substantially preventing any material from the tube **114** from escaping from the first end **106** of the adapter body **104**. As such, the adapter **102** and needle member **110** are operably configured to deliver the adhesive or other material through the internal channel and needle channel, respectively, out through an opening defined by the distal end **206** of the needle member **110**.

(34) In one embodiment, the body **104** is made from of a substantially rigid polymeric material, e.g., medium-density nylon, and may be of a monolithic structure forming the internal channel. The adapter **102** may beneficially include one or more longitudinally oriented spaced-apart flanges **118a-n** configured to facilitate the user in selectively coupling and uncoupling the adapter from the mouth **112** of the tube **114**, wherein “n” represents any number greater than one. The adapter **102** may also include a recessed internal wall surface **212** defining a threaded configuration configured to thread or couple with the threads on the mouth **112** of the tube **114**, typically in a watertight configuration. In preferred embodiments, only the lower section of the adapter **102** has the recessed internal wall surface **212**.

(35) Said another way, the adaptor **102** is configured to attach to a pre-filled standard 10 oz. size tube cartridge **114** that features an external threaded nipple/mouth **112** that is conventionally formed to accept a nozzle to emit contents of the tube cartridge **114**. The user may need to drill a hole or otherwise create an opening in the mouth **112** of the tube **114** to provide access to the contents of the tube cartridge **114**. The adaptor fitting **102** attaches to the nipple/mouth **112** and permits either the needle member **110** or cap **400** to selectively couple thereto, thereby preserving the contents of the tube **114** when not desired for use. When desired for use, the needle member **110**, preferably along with a caulk gun, enables the contents of the tube, e.g., adhesive, to travel through the adapter **102**, needle member **110**, out through the distal end **206**, and dispense into the void between the flooring material **300** and the subfloor **302** (until typically reaching another gap or opening created between the flooring material **300**—wherein any excess adhesive material is wiped clean). The resealable cap **400** is operably configured to seal, preferably hermetically, the second end **108** or internal channel **200** when the adapter **102** is not desired for use, thereby beneficially preventing the liquid adhesive from drying out. The second end **108** is also operably configured and proportionately sized and shaped to receive the injection needle **110**. In an exemplary embodiment, the injection needle **110** is a blunt end, non-sterile stainless steel, 12- or 14-gauge needle with an inert plastic or stainless-steel hub **116**. The needle is preferably of a rigid non-flexible material, e.g., stainless steel or PVC. Said another way, the needle member **110** may not be capable of bending under bending forces of approximately 51 bf. The needle member **110** has a blunt end at the terminal end of the needle member **110** to maintain an even distribution of the adhesive material and prevent puncturing of any flooring material. Said another way, the terminal end does not taper in diameter to form a point that most needle members are commonly formed in.

(36) Exemplary specifications of the injection needle **110** are an outer diameter of 0.083, inner diameter of 0.069, weight of 0.08 pounds, and length of 1.5 inches. FIG. 5 depicts several exemplary, yet preferred, lengths of the injection needle **110**, i.e., lengths of the hub/needle **500**, **504**, **508** and needle **502**, **506**, **510** of approximately 2.2, 1.7, 1.2 in. and 1.5, 1.0, 0.5 in., respectively. As such, the length of the needle **110** relative to the outer diameter, exemplified below, provides an advantageous slender configuration of the needle **110** that effectuates repeated and durable use of the needle **110** for entering and removing the needle from an opening between two pieces of flooring material **300**. In one embodiment, the slender configuration uniformly spans the entire length of the needle **110**. Said another way, the assembly **100** may have an overall hub-needle length **500** separating the first end **210** of the hub **116** and the distal end **206** of the needle member **110** and a needle member length **502** separating the second end **214** of the hub **116** and the distal end **206** of the needle member **110**, wherein the overall hub-needle length **500** may beneficially have a length variance ranging from 0.5-1.5 in. relative to the needle member length **502** to ensure needle member **110** integrity and effectiveness when utilized with the flooring material **300** and subfloor **302**.

(37) Additionally, the needle **110** may be beneficially formed in a linear configuration. The blunt end improves user safety, while the narrow diameter provides a precise and micro application of the liquid adhesive into tight or narrow spaces. The second end **108** is operably configured to receive injection needles **110** having any of the following dimensions:

(38) TABLE-US-00001 Inner Diameter Outer Diameter Gauge IN MM IN MM 14 0.069 1.77 0.083 2.11 15 0.059 1.51 0.072 1.83 16 0.053 1.35 0.065 1.65 18 0.038 0.99 0.049 1.27 19 0.033 0.79 0.043 1.06 20 0.025 0.64 0.036 0.91 21 0.022 0.56 0.033 0.82 22 0.020 0.46 0.028 0.72 23 0.016 0.41 0.025 0.64 25 0.012 0.30 0.020 0.51 26 0.011 0.26 0.019 0.46 27 0.009 0.22 0.016 0.41 30 0.006 0.15 0.012 0.31

(39) In one embodiment, the first end **106** of the body **104** is operably configured for attachment to a cartridge **114** containing the liquid adhesive. Beneficially, the assembly **100** may be reused multiple times, even after the cartridge **114** containing the liquid adhesive is fully used and empty. The cartridge **114** is simply replaced with a new one and the first end **106** of the assembly **100** is attached to the new cartridge **114**. For ease of use, the cartridge **114** containing the liquid adhesive may be received in a caulk gun which would provide for a faster and more efficient application of the liquid adhesive. The hub and needle member **110** may also be beneficially replaced depending on the application and/or wear of the needle member **110**.

(40) Said another way, the assembly **100** may include an adapter **102** with an adapter body **104** having a first end **106**, a second end **108** opposing the first end **106** of the adapter body **104**, a recessed internal wall surface **212** defining a threaded configuration operably configured to couple with a threaded tube **114**, and that defines an internal channel **200** spanning from the first end **106** of the adapter body **104** to the second end **108** of the adapter body **104**. The assembly **100** also includes a hub **116** selectively removably coupled to the adapter body **104**, having a first end **210** disposed in a directly adjacent configuration with the second end **108** of the adapter body **104**, having a second end **214** opposing the first end **210** of the hub **116**, and having a rigid needle member **110** coupled thereto and recessed within the hub **116**. The rigid needle member **110** can be seen projecting outwardly from the second end **214** of the hub **116** in a linear orientation, having a blunt distal end **206**, and defining a needle channel **204** spanning from the blunt distal end **206** to the first end **210** of the hub **116**, wherein the needle channel **206** is in fluid communication with the internal channel **200**.

(41) The present invention also discloses a micro injection method for the repair of floors where the flooring material **300** has become separated from the subfloor **302** and created a void between the flooring material **300** and the subfloor **302**. The method comprises an initial Step of determining a separated area of the flooring material **300**, i.e., a loose, hollow, or separated, broken flooring. The problem area(s) may be located using a tapping dowl and marked using chalk or tape. A next Step entails drilling a plurality of holes within the problem area(s). Any debris within the plurality of holes may be vacuumed out to beneficially facilitate a stronger or cleaner adhesive bond. The first end **106** of the adaptor **102** is then screwed onto the cartridge **114** containing the liquid adhesive. Next, the resealable cap **200** is unscrewed and removed from the adaptor body **104**. The injection needle **110** is then screwed onto and attached to the adaptor **102**. In accordance with a next Step, and using the caulk gun, the user slowly and gently injects the liquid adhesive from the cartridge **114** with the fully assembled adaptor **102** attached, into the plurality of holes drilled into the floor. Specifically, before the liquid adhesive is injected into the void, the cartridge **114** containing the liquid adhesive is placed in a caulk gun for greater ease of use and application of the liquid adhesive. A next Step entails stopping injection of the liquid adhesive followed by the Step of allowing the liquid adhesive to set. If needed, the resealable cap **200** is screwed onto and attached to the adaptor **102** to keep any remaining adhesive from drying out in the cartridge **114** for future use. Any excess adhesive is then wiped off the floor with a wet sponge and the area is allowed to dry for 24 hours. Any holes are then filled with grout or wood filler.

(42) Said another way, the present invention may be described as a method of repairing loose flooring material **300** relative to a subfloor **302** and may first start at step **600** and proceed to step **602** of providing an adapter body **104**, hub **116**, needle member **110**, and/or cap **400** with many of the above-referenced features. For example, the adapter body **104** may include a threaded configuration and an internal channel **200** therethrough, the hub **116** may include the rigid needle

member **110** coupled thereto, of a slender configuration, defining the needle channel **206** therethrough, and having the distal end **206** defining an opening from which adhesive material is emitted. The invention also includes providing the cap **400** that may be selectively removably couplable to the adapter body **104**.

(43) Next, step **604** includes coupling the threaded configuration of the adapter body **104** to a threaded configuration of a tube **114** filled with an adhesive material **306**. As part of this step **604** or after this step **604**, the process may include coupling the hub **116** to the adapter body in a watertight configuration. Before step **604**, the process may include opening the tube **114**, particularly the mouth **112** of the tube **114**, to provide the user access to the adhesive or other material within the tube **114**. As mentioned previously, this may be done by drilling a hole in the mouth of the tube **114**. When the hub **116** and needle member **110** are selectively attached to the adapter body **104**, the needle member **110** is preferably configured in a linear orientation. Said another way, the needle member **110** is oriented and/or parallel to the orientation of the tube **114**. When attached to the tube **114**, the internal channel **200** and the needle channel **204** may also be overlapping with or concentrically displaced relative to the upper opening **202** defined by the mouth **112**. In one embodiment, the process includes coupling the hub **116** to the adapter body **104** in directly adjacent configuration and, in some embodiments, coupling the hub **116** to the adapter body **104** by inserting the hub **116** within the recessed annular channel **216**.

(44) Next, as best exemplified by FIG. 3, step **606** includes inserting the rigid needle member **110** into an opening **304** adjacent to or between two pieces of flooring material **300** loosely coupled to a subfloor. Said another way, the flooring material **300** may be at least partially separated from the subfloor **302** such that when the user modulates the tube **114** to emit adhesive out of the needle member **110** it causes the adhesive to flow where the flooring material **300** is separated and ultimately up through the opening where the needle member **110** is inserted or another opening between the flooring material **300**. In one embodiment, the needle member **110** is inserted into the opening until reaching the subfloor **300** or other stopping point, whereby the rigidity, dimensions, and/or configuration of the needle member **110** prevents the needle member **110** from inadvertently breaking and prolongs the utilization thereof.

(45) Next, the process includes the step **608** of modulating the tube **114** (e.g., by using a piston on a caulk gun) to transport the adhesive material from the tube **114** through the internal channel **200**, through the needle channel **206**, out through the opening of the distal end **206** of the rigid needle member **110**, and between the flooring material **300** and subfloor **302**. When the adhesive material cures, this beneficially reseals or attaches the flooring material **300** to the subfloor **302**.

(46) Next, the process may include proceeding to step **610** of uncoupling the hub **116** from the adapter body adapter body and coupling the cap **400** to the adapter body **104** in a watertight configuration when the user desires to reuse the tube **114**. The cap **400** may couple and uncouple to the adapter body **104** using a tongue-and-groove configuration and in a hermetically sealed configuration (e.g., using a rubber seal). The process may terminate at step **612**.

(47) ces between a flooring material and subfloor.

Claims

1. A micro injection assembly for the repair of floors comprising: an adapter body **104** having a first end, a second end opposing the first end of the adapter body, a recessed internal wall surface defining a threaded configuration operably configured to couple with a threaded tube, and defining an internal channel spanning from the first end of the adapter body to the second end of the adapter body; a hub **116** selectively removably coupled to the adapter body, having a first end **210** disposed in a directly adjacent configuration with the second end of the adapter body **104**, having a second end opposing the first end of the hub, and having a rigid needle member **110** coupled thereto and recessed within the hub, the rigid needle member projecting outwardly from the second end of the

hub in a linear orientation, having a blunt distal end **206**, and defining a needle channel spanning from the blunt distal end to the first end of the hub, wherein the needle channel is in fluid communication with the internal channel; an overall hub-needle length separating the first end of the hub and the distal end of the needle member and a needle member length separating the second end of the hub and the distal end of the needle member, wherein the overall hub-needle length has a length variance ranging from 0.5-1.5 in. relative to the needle member length; and a cap selectively removably couplable to the adapter body, when the hub is not coupled to the adapter body, in a watertight configuration with the cap disposable in an overlapping configuration with the second end of the adapter body.

2. The micro injection assembly according to claim 1, wherein the adapter body further comprises: a lower section having the first end of the adapter body and the recessed internal wall surface; and an upper section adjacent to the lower section, having the second end of the adapter body, and tapered in diameter relative to a diameter of the lower section of the adapter body.

3. The micro injection assembly according to claim 2, wherein the upper section further comprises: a spout surrounded by a recessed annular channel, the recessed annular channel with the hub disposed therein.

4. The micro injection assembly according to claim 2, wherein the lower section and the upper section are of a monolithic structure.

5. The micro injection assembly according to claim 1, wherein the rigid needle member is of a slender configuration.

6. The micro injection assembly according to claim 5, wherein the rigid needle member ranges in dimension from 12-14 gauge.

7. The micro injection assembly according to claim 1, wherein the adapter body further comprises: a spout surrounded by a recessed annular channel, the recessed annular channel with the hub disposed therein.

8. A method of repairing loose flooring comprising the steps of: providing an adapter body with a threaded configuration and an internal channel therethrough, a hub with a rigid needle member coupled thereto, of a slender configuration, having, defining a needle channel therethrough, and having a distal end defining an opening, and a cap, further comprising an overall hub-needle length separating a first end of the hub and a distal end of the needle member and a needle member length separating a second end of the hub where the rigid needle member protrudes and the distal end of the needle member, wherein the overall hub-needle length has a length variance ranging from 0.5-1.5 in. relative to the needle member length; coupling the threaded configuration of the adapter body to a threaded configuration of a tube filled with an adhesive material and coupling the hub to the adapter body in a watertight configuration; inserting the rigid needle member into an opening adjacent to a piece of flooring material loosely coupled to a subfloor, wherein the rigid needle member ranges in dimension from 12-14 gauge; modulating the tube to transport the adhesive material from the tube through the internal channel, through the needle channel, out through the opening of the distal end of the rigid needle member, and between the flooring material and subfloor; and uncoupling the hub from the adapter body adapter body and coupling the cap to the adapter body in a watertight configuration.

9. The method according to claim 8, further comprising: coupling the cap to the adapter body in a hermetically sealed configuration.

10. The method according to claim 8, further comprising: coupling the hub to the adapter body in directly adjacent configuration.

11. The method according to claim 8, further comprising: providing the adapter body with a spout surrounded by a recessed annular channel; and coupling the hub to the adapter body until inserted within the recessed annular channel.

12. The method according to claim 11, further comprising: threadedly coupling the hub to the adapter body.

13. The method according to claim 8, further comprising: providing the rigid needle member projecting from the hub in a linear orientation and having a blunt distal end.

14. The method according to claim 8, further comprising: providing the adapter body with a lower section having a first end of the adapter body and a recessed internal wall surface and an upper section adjacent to the lower section, having a second end of the adapter body located thereon, and tapered in diameter relative to a diameter of the lower section of the adapter body; and coupling the recessed internal wall surface, with the threaded configuration thereon, to the adapter body to the tube.
